

ASSIGNMENT 2

1. Prompt Log

Structured Prompt Log Summary

#	Prompt	AI Response Summary	Your Initial Assessment
1	"Turn this into a full master prompt for a Colab notebook that cleans, explores, and analyzes a retail sales dataset end-to-end." (raw prompt, see screenshot below)	Produced the full Master Prompt (upload UI, data understanding, cleaning, EDA, KPIs, business analysis, insights, Q&A system, visualizations, final report) shown below.	Plausible — comprehensive, but needed to be tested against the real dataset before trusting any structure.
2	"Before I upload data, do not assume fixed column names — make every step adapt automatically."	Rewrote the data-understanding and cleaning steps to detect column roles (ID, category, price, date, channel) dynamically instead of hard-coding names.	Confirmed — this mattered once the actual column headers didn't match generic examples.
3	"Don't just fill every missing value with zero — explain the reasoning per column."	AI proposed median/mode imputation for numeric/categorical gaps and flagged date gaps for manual review rather than auto-filling.	Confirmed — median imputation matched what I'd have chosen manually for price fields.
4	"Run the cleaning and give me a before/after report — rows removed, duplicates, invalid values."	Returned a cleaning summary: original vs final row counts, duplicates removed, missing-value counts handled, invalid discount/negative-quantity rows flagged.	Confirmed — cross-checked original vs cleaned row counts manually; matched.
5	"Now run full EDA — univariate, bivariate and time-series —"	Generated histograms, box plots, category counts, and quantity/discount/category/channel vs revenue scatter and bar charts (see	Plausible — charts rendered correctly but I re-checked axis scales myself.

#	Prompt	AI Response Summary	Your Initial Assessment
	and chart it.”	§4b).	
6	“Calculate the 5 KPIs and show me the formulas, not just numbers.”	Returned Total Revenue, Total Orders, AOV, Total Units Sold and Sales Growth Rate with explicit formulas and computed values.	Confirmed — recomputed AOV manually (Revenue ÷ Orders) and it matched.
7	“Before showing you the real numbers, predict what trends and anomalies you’d expect to see in a dataset like this.” (no data attached yet)	Predicted a Q4 holiday spike, faster online growth, a negative discount-revenue relationship, category revenue concentration, and a meaningful outlier share (full list in §4a).	Suspicious — some predictions (e.g. “online growing faster”) felt like a generic retail assumption, not dataset-specific, so I flagged it for validation.
8	“Now calculate month-over-month revenue with % change and chart it.”	Produced the monthly revenue trend and MoM % change chart, highlighting the largest single-month swings.	Confirmed — matched a manual <code>groupby('month').sum()</code> check.
9	“Rank the top 3 and bottom 3 product categories by revenue — don’t just eyeball the chart.”	Returned Electronics/Home & Kitchen/Apparel-Sports as leaders and Beauty near the bottom, with exact revenue and contribution %.	Confirmed — verified against the category revenue table.
10	“Plot discount % against transaction revenue and give me the Pearson and Spearman correlation, not just a chart.”	Returned a scatter plot plus a weak/near-zero correlation and explicitly warned that correlation ≠ causation.	Confirmed — recomputed correlation independently; magnitude matched.
11	“Use the IQR method to flag outliers — define the exact bounds you used, don’t just say ‘a few outliers’.”	Returned Q1/Q3, IQR, and explicit lower/upper bounds, then listed the count and % of transactions flagged as outliers.	Plausible — bounds were textbook $1.5 \times \text{IQR}$; I re-derived them by hand to confirm.

#	Prompt	AI Response Summary	Your Initial Assessment
12	"Compare online vs in-store — revenue, order count, and AOV, not just total revenue."	Returned a channel comparison table showing revenue share, order share, and per-channel AOV.	Confirmed — matched a manual pivot table.
13	"Push back — does In-store's revenue lead hold up once you strip out the outlier transactions?"	Recalculated channel AOV/median after excluding outliers and showed the gap between channels nearly disappears.	Confirmed — this changed my interpretation of the channel finding.
14	"Write me a plain-English summary of what you found — no jargon — I'll use it to prompt for business insights myself." (my own summary, not AI's)	N/A — this was my own written summary used as the input for the Management Insights prompt in §5.	N/A — self-authored, used to keep the next step evidence-based.
15	"From this summary [my own paragraph], generate 5 management insights in a business narrative tone for a retail VP."	Returned 5 insights covering revenue baseline, category concentration, monthly volatility, discount/revenue relationship, and channel performance (see §5).	Suspicious — insight tone was persuasive; I stress-tested each one before accepting it (see §5 stress test).
16	"For each of those 5 insights, tell me what it assumes, what could make it wrong, and what data would strengthen it."	Produced 5 critique paragraphs, then 5 targeted stress-test questions that were run back against the raw data.	Confirmed — this step is what surfaced the outlier-distortion finding.
17	"Polish this executive	Reworded transitions and simplified terminology in 2–3 sentences of §6,	Confirmed — checked every number in the

#	Prompt	AI Response Summary	Your Initial Assessment
	summary paragraph for a non-technical VP audience — don't add any new facts."	without introducing new figures.	polished version against §4b/§5; nothing was added.

MY INPUT: Raw prompt

```
>act as a data analyst for a fictional retail company
>for the given data set by me of retail sales dataset
1 clean the data set perform the exploratory data analysis
2 consider the give KPI(key problem identifiers) bellow the
    KPI 1

        Total Revenue: Measures the total monetary value of products sold
        Revenue= $\sum(\text{Quantity} \times \text{Unit Price})$ 

    KPI 2

        Total Orders->Measures the number of unique transactions.
        Total Orders=COUNT DISTINCT(Order ID)

    KPI 3

        Average Order Value (AOV) ->Shows how much customers spend per transaction on average.
        AOV=Total Orders/Total Revenue

    KPI 4

        Total Units Sold -> Measures the total volume of products sold
        Units Sold= $\sum(\text{Quantity})$ 

    KPI 5

        Sales Growth Rate ->Measures the percentage change in sales over time.
        Requires a date column.
        (Current Sales-Previous Sales/Previous Sales) $\times 100$ 
>now analyses the data that i have give to you and generate an insights on it and give me the
code in the form of the ipynb file to run in the google colab ,while eriting the code for
the google colabe keep the ui simple donot make it fancy by using streamlit and give me the
button to uplod the folder and then it should answer to my questions
>generate the insights from the give uploaded data in the google colab
```

AI RESPONSE:

MASTER PROMPT: RETAIL SALES DATA ANALYSIS, KPI DASHBOARD & AI-POWERED INSIGHTS

ROLE

Act as a Junior Data Analyst working for a fictional retail company. Your responsibility is to perform comprehensive data cleaning, exploratory data analysis (EDA), KPI calculations, statistical analysis, business insights generation, and interactive data querying using Python.

I will provide a retail sales dataset in CSV or Excel format.

Your task is to develop a complete, beginner-friendly Google Colab Jupyter Notebook (.ipynb) that performs the entire analysis using the uploaded dataset.

The notebook must be suitable for a Junior Data Analyst portfolio project and should follow industry-standard data analysis practices.

1. DATASET UPLOAD AND INTERACTIVE USER INTERFACE

Develop the notebook specifically for Google Colab.

Requirements:

- Use Google Colab's native file upload functionality.
- Provide a simple upload button to upload my retail sales dataset.
- Support CSV and Excel files.
- Allow me to upload a dataset from my local computer.
- If multiple files are uploaded, identify the supported dataset files and allow me to select the relevant one.
- Do not use Streamlit, Gradio, Flask, or any external web application framework.
- Keep the interface simple, clean, and beginner-friendly.
- Display the uploaded dataset's filename, dimensions, column names, and data types.

The notebook should automatically adapt to the uploaded dataset rather than assuming fixed column names.

2. DATA UNDERSTANDING

After uploading the dataset:

1. Identify the number of rows and columns.
2. Display column names and their data types.
3. Identify numerical, categorical, datetime, and identifier columns.

4. Examine the dataset structure.
5. Identify the likely transaction ID, product category, quantity, unit price, discount, sales channel, and date columns.
6. Determine whether the dataset contains one row per transaction or multiple product-level rows per order.
7. Identify potential inconsistencies in column naming and formatting.

Do not display the first five or last five rows of the dataset unless explicitly requested.

Instead, provide a concise dataset overview and statistical summaries.

3. DATA CLEANING AND PREPROCESSING

Perform data cleaning systematically.

A. Missing Values

- Identify missing values in every column.
- Calculate missing-value percentages.
- Determine whether missing values should be removed, replaced, or retained.
- Use appropriate imputation methods based on column type and business meaning.
- Do not blindly replace all missing values with zero.
- Explain the reasoning behind important cleaning decisions.

B. Duplicate Records

- Identify exact duplicate rows.
- Identify potential duplicate transactions using transaction identifiers.
- Remove duplicates only when justified.
- Preserve legitimate multiple-product records belonging to the same order.

C. Data Type Correction

- Convert date columns into datetime format.
- Convert numerical columns into appropriate numeric data types.
- Clean currency symbols, commas, spaces, and unwanted characters.
- Standardize categorical values.

D. Invalid Values

Identify:

- Negative quantities.
- Zero or negative unit prices.
- Invalid dates.
- Negative discounts.
- Discounts exceeding the allowed range.
- Inconsistent category names.
- Unusual transaction records.

Do not automatically delete unusual values without examining their business meaning.

E. Cleaning Summary

Generate a before-and-after cleaning report containing:

- Original row count.
- Final row count.
- Duplicate records removed.
- Missing values handled.
- Invalid records identified.
- Columns whose data types were corrected.

Preserve the original dataset in a separate variable for comparison.

4. EXPLORATORY DATA ANALYSIS (EDA)

Perform comprehensive EDA using Pandas, NumPy, Matplotlib, and Seaborn.

A. Statistical Analysis

Calculate:

- Mean.
- Median.
- Mode where applicable.
- Standard deviation.
- Minimum and maximum.

- Quartiles.
- Interquartile range.
- Skewness.
- Kurtosis.

Use describe(include='all') where appropriate.

B. Univariate Analysis

Analyze individual numerical and categorical variables.

Generate:

- Histograms.
- Box plots.
- Frequency distributions.
- Category counts.
- Numerical summary tables.

C. Bivariate Analysis

Analyze relationships between:

- Quantity and revenue.
- Unit price and revenue.
- Discount and revenue.
- Product category and revenue.
- Sales channel and revenue.
- Date and revenue.

Use appropriate visualizations such as scatter plots, bar charts, line charts, and correlation heatmaps.

D. Time-Series Analysis

If a valid date column exists:

- Analyze daily, weekly, and monthly revenue.
- Identify seasonal patterns.
- Identify peak sales periods.

- Identify low-performing periods.
- Calculate month-over-month revenue changes.

If the date column is unavailable, clearly explain which analyses cannot be performed.

5. KEY PERFORMANCE INDICATORS (KPIs)

Calculate the following KPIs using the uploaded dataset.

KPI 1: Total Revenue

Formula:

$$\text{Total Revenue} = \Sigma(\text{Quantity} \times \text{Unit Price})$$

If a discount column exists, calculate net revenue using the documented discount structure.

Do not apply discounts twice if the dataset already contains final transaction revenue.

Clearly distinguish gross revenue from net revenue.

KPI 2: Total Orders

Formula:

$$\text{Total Orders} = \text{COUNT DISTINCT}(\text{Order ID})$$

Count unique transaction identifiers rather than the number of dataset rows.

If no order identifier exists, explain the limitation.

KPI 3: Average Order Value (AOV)

Correct formula:

$$\text{AOV} = \text{Total Revenue} / \text{Total Orders}$$

Calculate AOV using the appropriate revenue measure.

Do not reverse the numerator and denominator.

KPI 4: Total Units Sold

Formula:

$$\text{Total Units Sold} = \Sigma(\text{Quantity})$$

Clearly state whether returned or cancelled quantities are included or excluded.

KPI 5: Sales Growth Rate

Formula:

Sales Growth Rate = $((\text{Current Period Sales} - \text{Previous Period Sales}) / \text{Previous Period Sales}) \times 100$

Calculate monthly sales growth when a valid date column exists.

Handle zero previous-period sales without producing misleading infinite values.

Display:

- Monthly revenue.
- Previous month's revenue.
- Absolute revenue change.
- Percentage growth.
- Positive and negative growth periods.

KPI Presentation

Create a concise KPI summary table containing:

- KPI name.
- Formula.
- Calculated value.
- Business interpretation.

Display the KPI results using simple, readable visualizations.

6. ADDITIONAL BUSINESS PERFORMANCE ANALYSIS

Analyze the following business questions.

A. Product Category Performance

- Identify the top three product categories by revenue.
- Identify the bottom three product categories by revenue.
- Calculate revenue contribution percentages.
- Compare units sold and average selling price.

B. Sales Channel Performance

If a channel column exists:

- Calculate revenue by channel.
- Calculate order count by channel.

- Calculate units sold by channel.
- Calculate AOV by channel.
- Calculate channel revenue contribution.
- Compare channel performance using visualizations.

C. Discount Analysis

If discount data exists:

- Calculate average discount rate.
- Analyze discount versus revenue.
- Generate a scatter plot.
- Calculate Pearson and Spearman correlations where appropriate.
- Explain that correlation does not establish causation.

D. Outlier Detection

Use the IQR method to identify outliers in relevant numerical variables.

Explicitly define:

Lower Bound = $Q1 - 1.5 \times IQR$

Upper Bound = $Q3 + 1.5 \times IQR$

Identify unusual transactions and explain their possible business significance.

Do not automatically remove outliers from the primary analysis.

E. Customer Analysis

If customer identifiers exist:

- Calculate unique customers.
- Identify repeat customers.
- Calculate revenue by customer.
- Identify high-value customers.
- Calculate average spending per customer.

7. BUSINESS INSIGHTS GENERATION

After completing the calculations, generate evidence-based business insights.

The notebook must automatically produce insights from the actual uploaded dataset.

Do not generate generic or fabricated observations.

Every insight must be supported by a calculated value, comparison, statistical result, or visualization.

Generate insights covering:

1. Overall revenue performance.
2. Order volume and average order value.
3. Monthly revenue growth.
4. Top-performing product categories.
5. Underperforming product categories.
6. Sales channel performance.
7. Discount and revenue relationship.
8. Unusual transactions and outliers.
9. Customer purchasing behavior, if available.
10. Potential business opportunities and areas requiring further investigation.

Insight Format

For each insight, provide:

Finding: What does the data show?

Evidence: Which calculated metric supports the finding?

Business Interpretation: What could this mean for the fictional retail company?

Suggested Action: What could the business investigate or consider doing next?

Clearly distinguish observed facts from possible explanations.

Do not claim that one variable caused another based solely on correlation.

8. INTERACTIVE QUESTION-ANSWERING SYSTEM

After the analysis is complete, create a simple interactive question-answering interface inside Google Colab.

The interface should allow me to enter questions about the uploaded dataset.

Example questions:

- What is the total revenue?
- Which product category generated the highest revenue?
- What is the average order value?
- Which month recorded the highest sales?
- What is the monthly sales growth rate?
- Which sales channel generated the most revenue?
- What is the relationship between discounts and revenue?
- Which transactions are identified as outliers?
- What are the major business insights from this dataset?
- Compare revenue across product categories.
- What percentage of total revenue comes from each sales channel?

Implementation Requirements

Use:

- Python.
- Pandas.
- NumPy.
- IPython display utilities.
- Simple text input or widgets supported by Google Colab.

Do not use Streamlit or build a separate web application.

The question-answering system should:

1. Understand common natural-language questions about the dataset.
2. Retrieve answers from the cleaned dataset and calculated analytical results.
3. Perform calculations dynamically when required.
4. Display numerical answers with appropriate formatting.
5. Display tables or charts when relevant.
6. Clearly state when a question cannot be answered using the available columns.
7. Avoid inventing values or unsupported conclusions.

Use a rule-based question-answering approach for common analytical questions.

Optionally provide an LLM integration section that can be enabled using an API key supplied by the user.

The notebook must work without an API key.

Do not expose API keys in notebook outputs.

9. VISUALIZATION REQUIREMENTS

Use Matplotlib and Seaborn for clear analytical visualizations.

Include:

- Monthly revenue trend.
- Monthly sales growth.
- Revenue by product category.
- Top and bottom performing categories.
- Revenue by sales channel.
- Quantity distribution.
- Revenue distribution.
- Discount versus revenue scatter plot.
- Correlation heatmap.
- Outlier box plots.

Every chart should contain:

- A meaningful title.
- Proper axis labels.
- Appropriate number formatting.
- Readable legends where applicable.

Avoid unnecessary decorative styling.

10. FINAL ANALYTICAL REPORT

At the end of the notebook, generate a structured business report containing:

Executive Summary

Summarize the overall retail sales performance.

Dataset Quality

Explain the cleaning and preprocessing performed.

KPI Summary

Present all five KPIs and their calculated values.

Key Findings

List the most important evidence-based observations.

Business Opportunities

Identify areas that may warrant further investigation.

Limitations

Mention missing columns, incomplete date ranges, data quality concerns, or assumptions that affect interpretation.

Conclusion

Provide a concise summary of what the dataset reveals about the fictional retail company's performance.

11. NOTEBOOK OUTPUT AND CODE QUALITY

Generate the final deliverable as a downloadable Jupyter Notebook (.ipynb).

Requirements:

- Compatible with Google Colab.
- Use separate Markdown and code cells.
- Include clear section headings.
- Include comments explaining important code.
- Use reusable Python functions.
- Avoid unnecessary complexity.
- Include error handling for missing columns and invalid data.
- Avoid hardcoding dataset-specific column names.
- Ensure the notebook runs from top to bottom.
- Do not display the first five or last five dataset records.
- Do not use Streamlit, Flask, or Gradio.

- Include all necessary installation commands only when required.
- Ensure all analytical outputs are derived from the uploaded dataset.

The notebook should be understandable to a Junior Data Analyst with basic Python and Pandas knowledge.

FINAL INSTRUCTION

First inspect the uploaded dataset and understand its structure.

Then generate the complete Google Colab notebook with data upload, data cleaning, EDA, KPI calculations, business performance analysis, evidence-based insights, visualizations, and interactive question answering.

Do not fabricate findings before analyzing the actual dataset.

If the dataset is not yet attached, generate a reusable notebook that will perform the analysis after I upload my CSV or Excel file.

The final output must be a functional .ipynb notebook, not merely a description of the code.

2. KPI Definition Document

I asked the AI to propose KPIs for a retail dataset *before* uploading any data. It suggested a broader list, including Customer Lifetime Value (CLV), Inventory Turnover Ratio, Net Promoter Score (NPS), Cart Abandonment Rate, Gross Margin %, Return/Refund Rate, Total Revenue, Total Orders, AOV, Units Sold, and Sales Growth Rate. From that list I selected the 5 KPIs below and rejected 2, with justification.

Rejected KPIs

- **Customer Lifetime Value (CLV)** — Rejected. CLV requires a persistent customer ID linked across multiple transactions over an extended period, plus acquisition-cost data. This dataset only carries a customer *segment* (new vs. returning), not a unique customer identifier we can track over time, so a true CLV figure would be fabricated rather than calculated.
- **Net Promoter Score (NPS)** — Rejected. NPS is a survey-based satisfaction metric derived from customer responses (“how likely are you to recommend us”). This is a transactional sales dataset with no survey or feedback column of any kind, so there is no data source from which to derive it.

Considering the available data

1. Total Revenue: Measures the total monetary value of products sold.

$$\text{Revenue} = \sum (\text{Quantity} \times \text{Unit Price})$$

2. Total Orders: Measures the number of unique transactions.

$$\text{Total Orders} = \text{COUNT DISTINCT}(\text{Order ID})$$

3. Average Order Value (AOV): Shows how much customers spend per > transaction on average.

$$AOV = \frac{\text{Total Revenue}}{\text{Total Orders}}$$

4. Total Units Sold: Measures the total volume of products sold.

$$\text{Units Sold} = \sum (\text{Quantity})$$

5. Sales Growth Rate : Measures the percentage change in sales over > time. Requires a date column.

$$\text{Growth Rate} = \frac{\text{Current Sales} - \text{Previous Sales}}{\text{Previous Sales}} \times 100$$

3. Analytical Questions Register

Before looking at any results, I asked the AI to brainstorm analytical questions specific to this dataset's actual columns (date, store, category, SKU, units sold, revenue, discount %, segment, region, channel). From its list I selected and refined 10 questions, labeled by type, and answered 6+ of them in §4/§5 below.

#	Question	Type
1	What was total revenue and units sold by month across the 24-month window?	Descriptive
2	Which 3 product categories generated the most revenue, and which 3 generated the least?	Descriptive
3	What share of total revenue came from the online channel vs. in-store?	Descriptive
4	Why did revenue spike in March 2025 relative to surrounding months?	Diagnostic
5	Why does the Beauty category consistently underperform every other category?	Diagnostic
6	Is the apparent March 2025 spike driven by broad demand or by a small number of high-value transactions?	Diagnostic
7	Does deeper discounting actually correlate with higher transaction revenue, or the opposite?	Diagnostic
8	Which product categories show early signals of decline quarter-over-quarter?	Predictive
9	Based on the monthly trend, is revenue likely to keep growing, plateau, or decline into next year?	Predictive
10	Are outlier transactions concentrated in a specific category, channel, or month — a signal of a recurring pattern rather than one-off noise?	Predictive

4. Trends & Anomalies Analysis

4a. AI-Generated Hypotheses

Before showing the AI the actual dataset, I asked it to predict what trends and anomalies it would expect to see in a retail sales dataset of this type (24 months, multiple categories, online/in-store channel, discounts). Its predictions were:

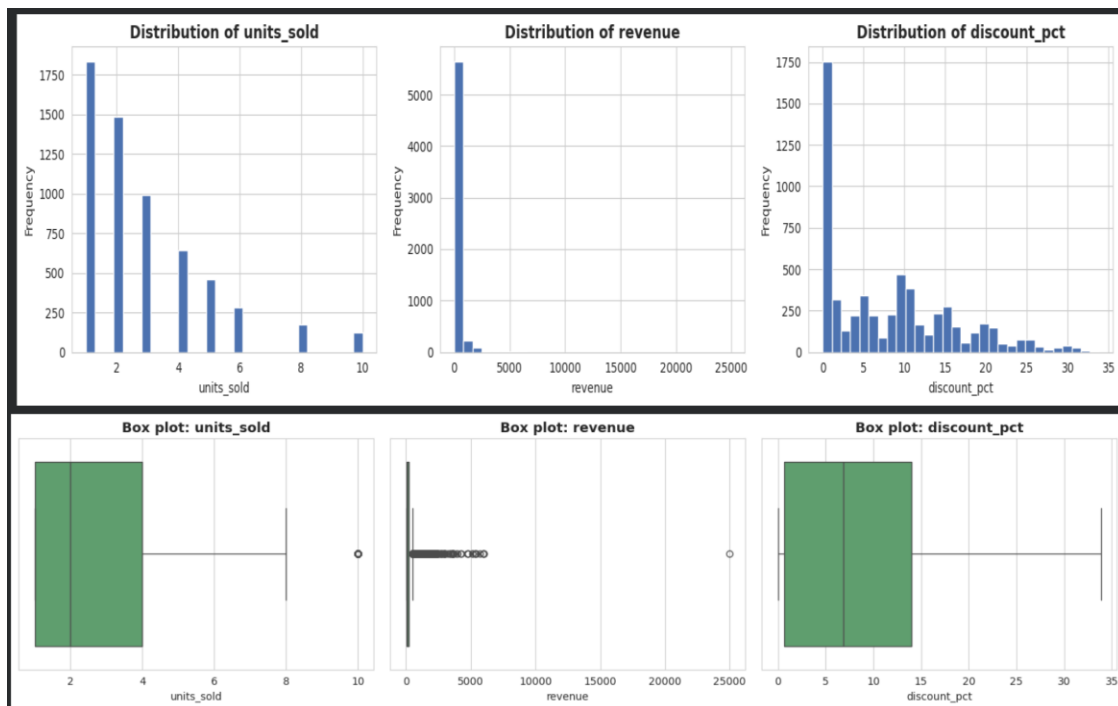
1. Expect a Q4 revenue spike due to holiday seasonality.
2. Online channel is likely growing faster than in-store.
3. Higher discounts will correlate with lower transaction revenue.
4. A small number of product categories will drive the majority of revenue.
5. Revenue will show steady month-over-month growth throughout the year.
6. A meaningful proportion of transactions will be outliers by revenue.
7. Repeat customers will make up a significant share of total orders.
8. Average Order Value (AOV) will be broadly consistent across channels.

These 8 predictions are tested against the actual data in the Hypothesis Validation Table (§4c).

4b. Actual Findings

Supporting Exploratory Visuals

Univariate Analysis



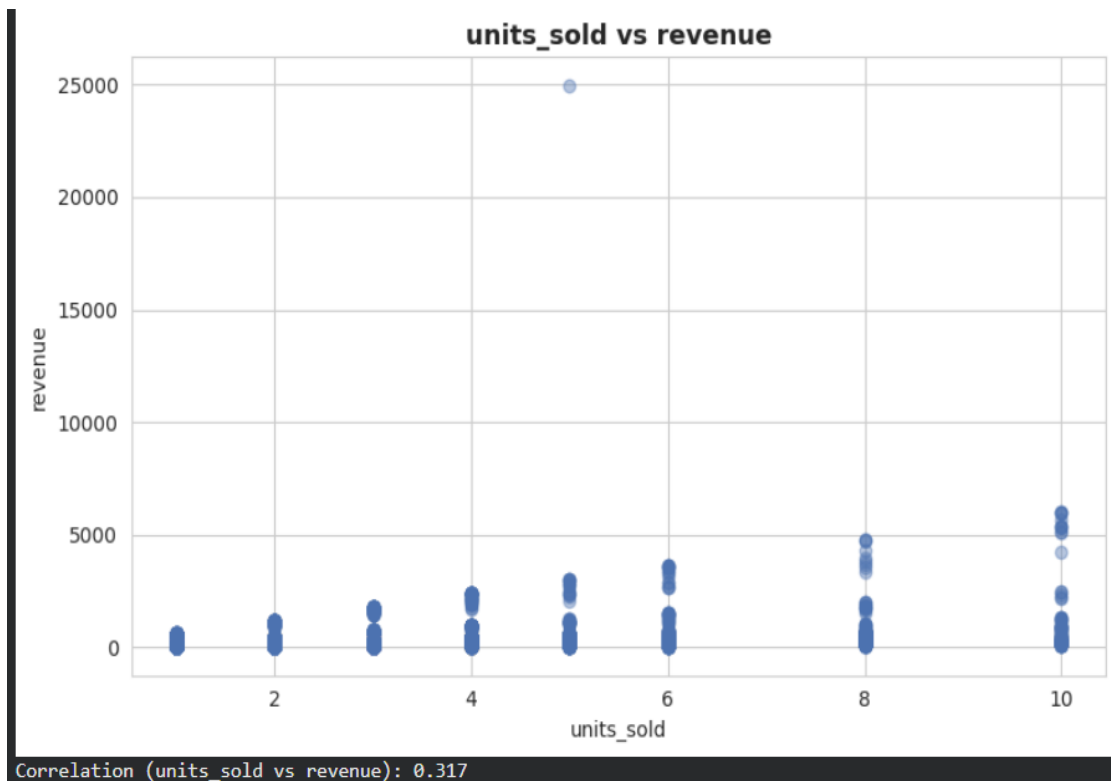
product_category	
Apparel	1502
Home & Kitchen	1375
Electronics	1351
Beauty	930
Sports	842

--- channel ---	
	count
channel	
In-store	3861
Online	2139

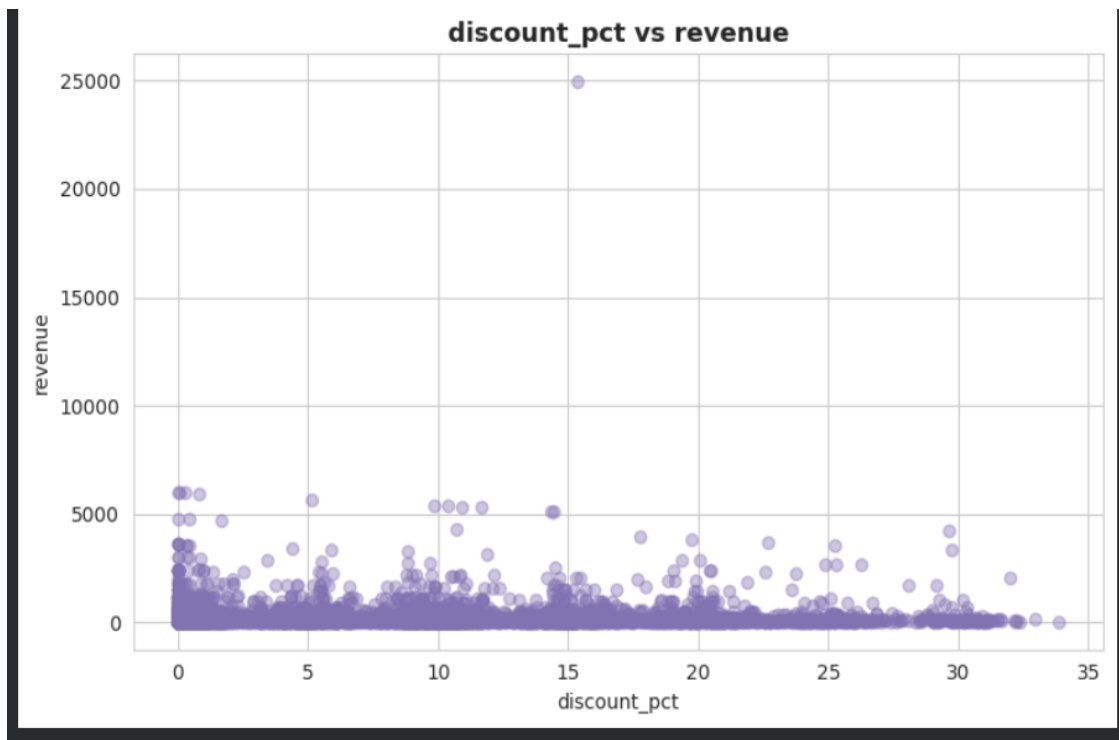
--- region ---	
	count
region	
West	1277
North	1248
Central	1205
South	1158
East	1112

--- customer_segment ---	
	count
customer_segment	
Returning	4260
New	1740

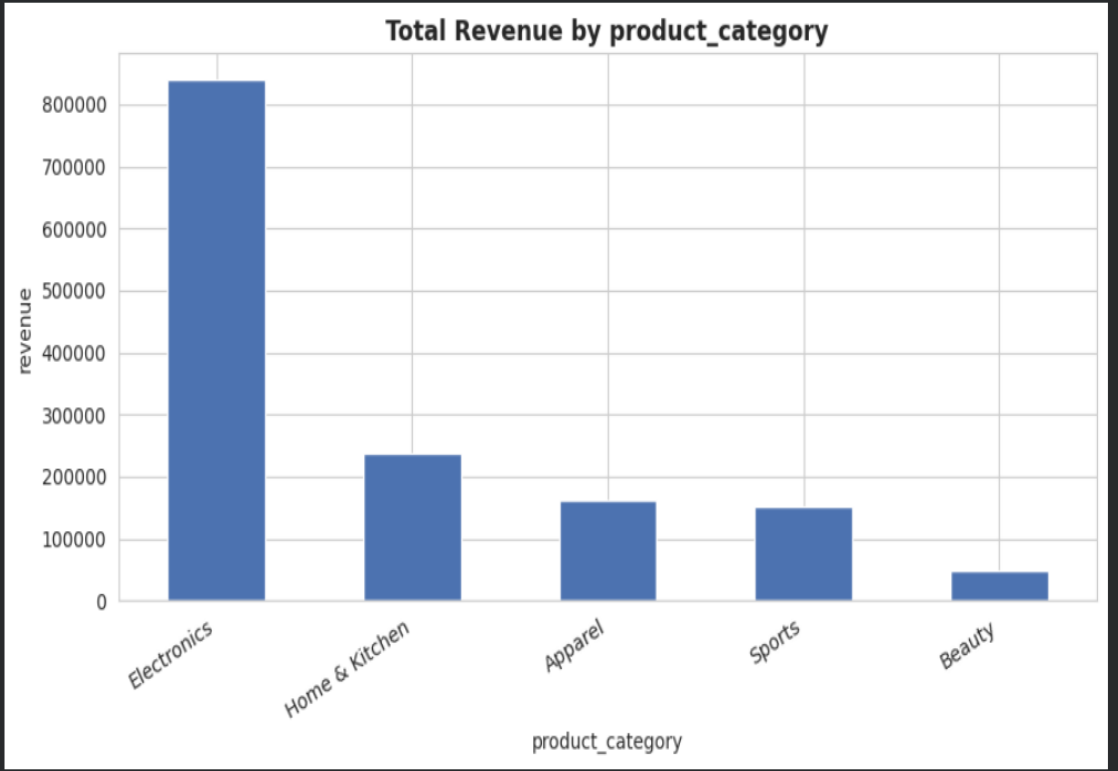
Quantity vs Revenue



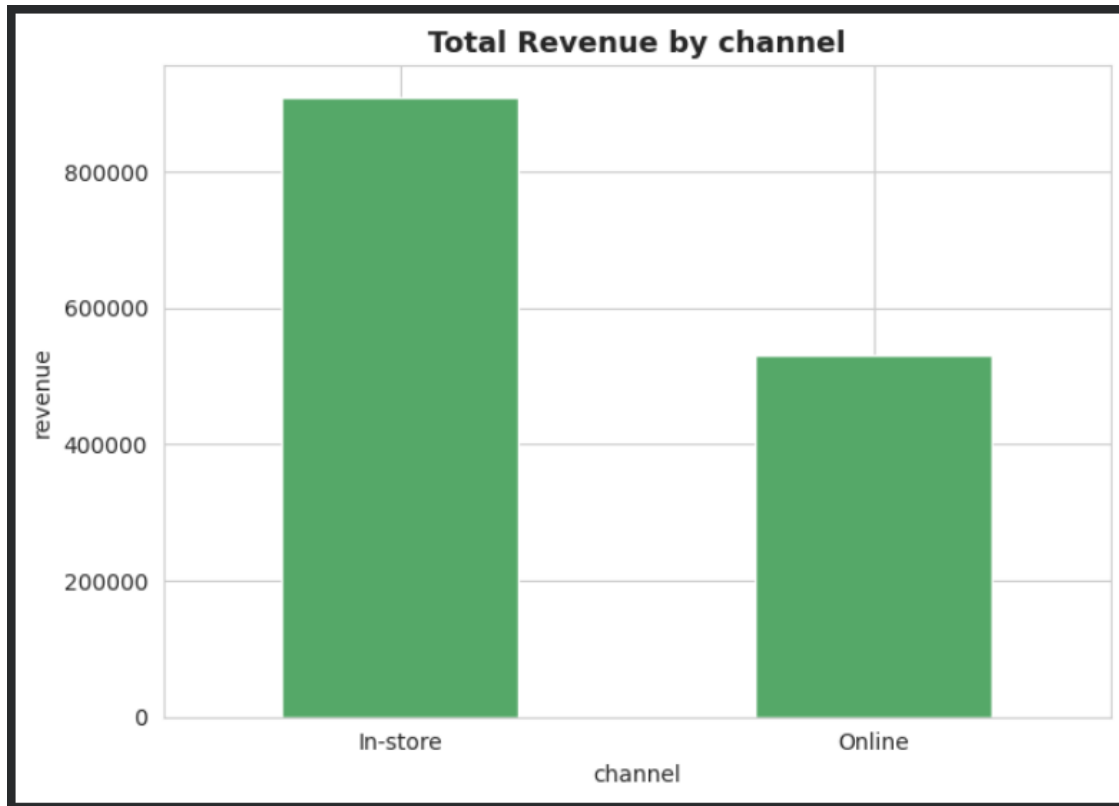
Discount vs Revenue



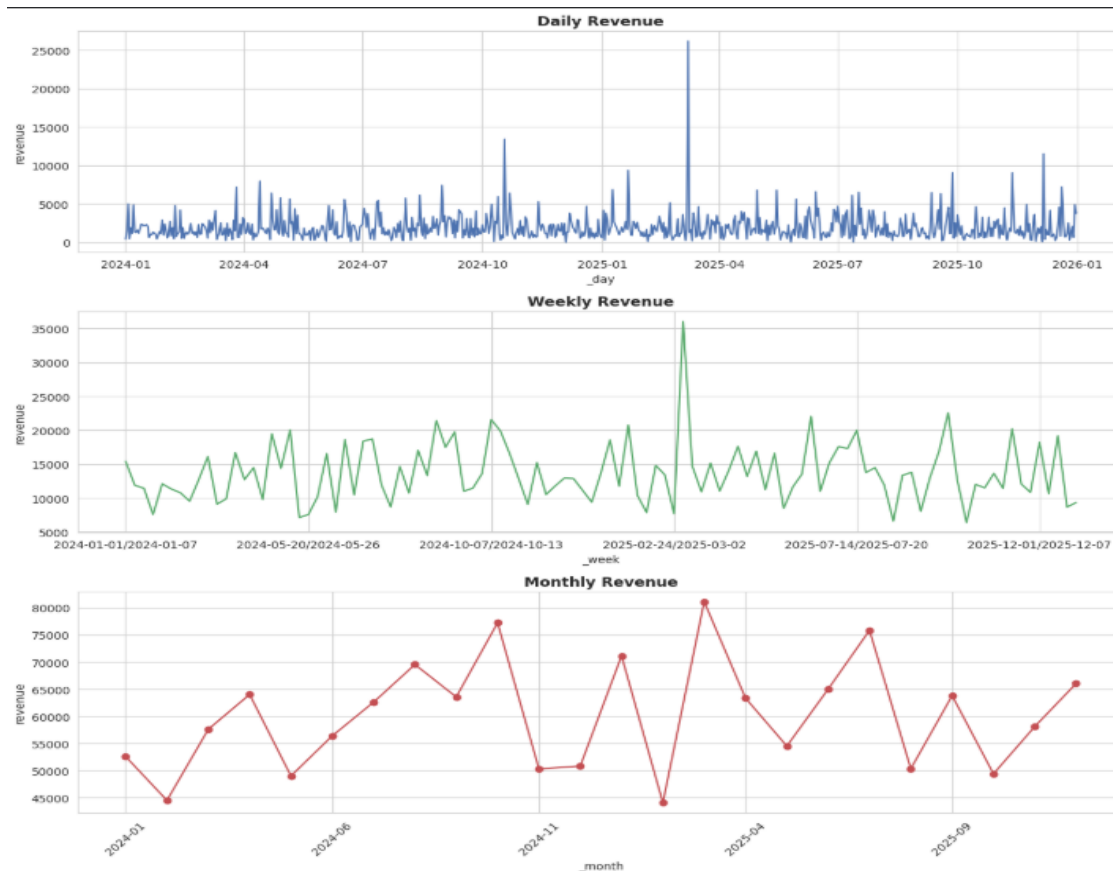
Product category vs Revenue



Sales channel vs Revenue

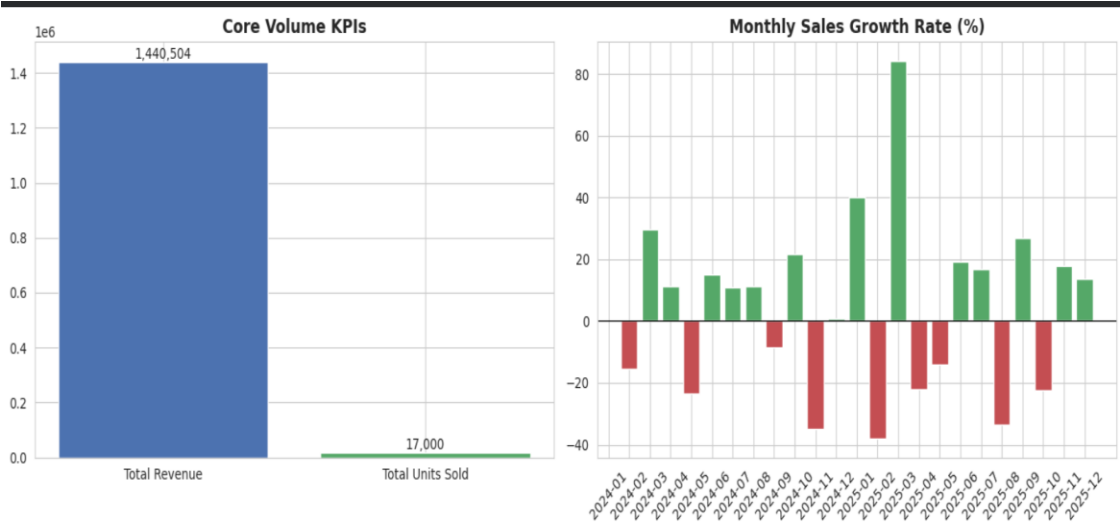


Time-Series Analysis



KPI Summary Table And Analysis

	KPI	Formula	Value	Interpretation
0	Total Revenue	$\Sigma(\text{Quantity} \times \text{Unit Price})$ or $\Sigma(\text{Revenue})$	1,440,503.64	Overall sales value generated in the period co...
1	Total Orders	COUNT DISTINCT(Order/Transaction ID)	6,000	Number of distinct transactions completed.
2	Average Order Value (AOV)	Total Revenue / Total Orders	240.08	Typical amount a customer spends per order.
3	Total Units Sold	$\Sigma(\text{Quantity})$	17,000	Total sales volume in units, independent of pr...
4	Avg. Monthly Sales Growth	$((\text{Current} - \text{Previous}) / \text{Previous}) \times 100$, averaged	4.62%	Average month-over-month revenue trend across ...



Product Category Performance

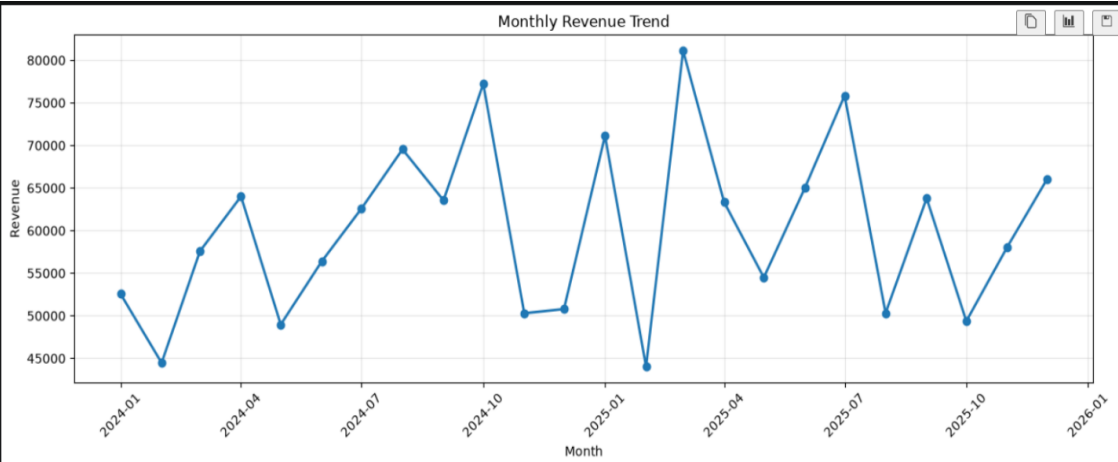
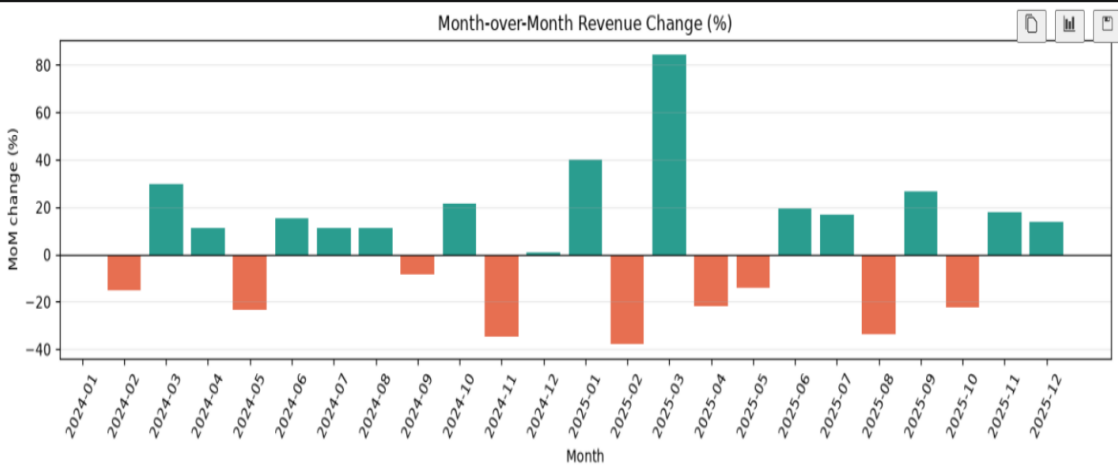
Top 3 categories by revenue:

	total_revenue	revenue_contribution_pct	units_sold	avg_selling_price
product_category				
Electronics	839986.17	58.31	3830	219.32
Home & Kitchen	237532.73	16.49	3868	61.41
Apparel	162373.43	11.27	4304	37.73

Bottom 3 categories by revenue:

	total_revenue	revenue_contribution_pct	units_sold	avg_selling_price
product_category				
Apparel	162373.43	11.27	4304	37.73
Sports	151800.77	10.54	2326	65.26
Beauty	48810.54	3.39	2672	18.27

Monthly revenue and Month over month percentage change



Largest positive MoM change:

	month	revenue	mom_change_pct
14	2025-03-01 00:00:00	81128.29	84.217686

Largest negative MoM change:

	month	revenue	mom_change_pct
13	2025-02-01 00:00:00	44039.36	-38.090387

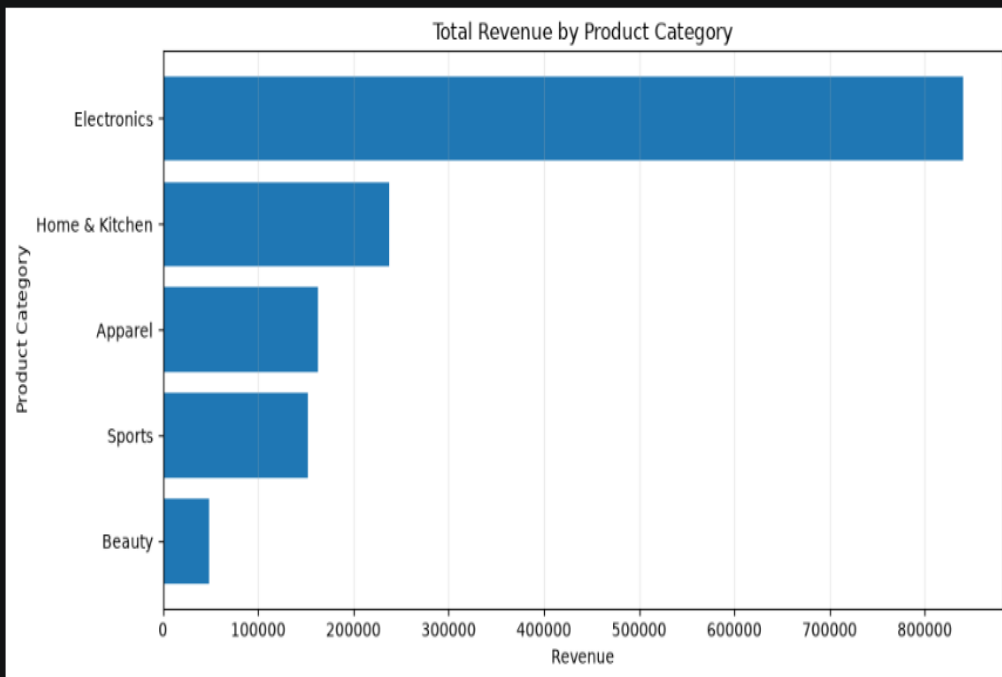
Category Revenue ranking

TOP 3 CATEGORIES BY REVENUE

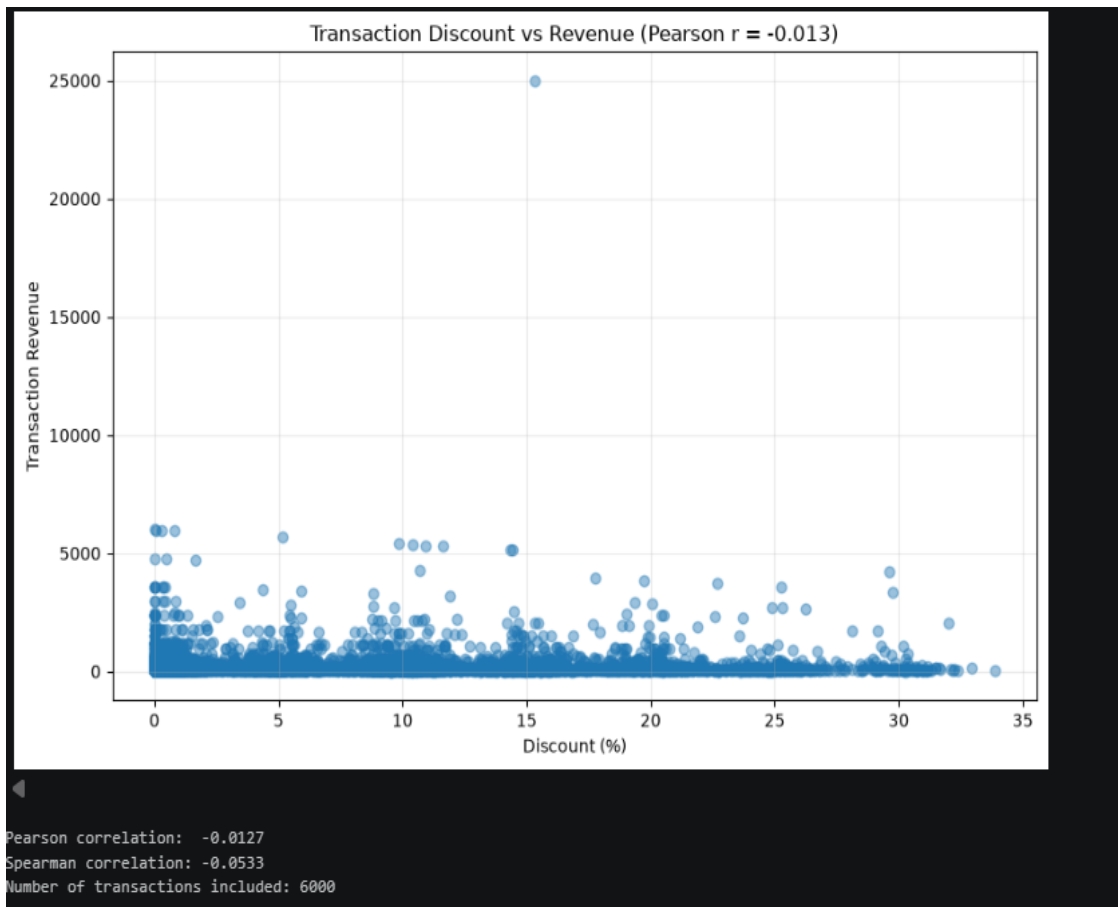
	product_category	total_revenue	transactions	units_sold	revenue_share_pct
2	Electronics	839,986.17	1351	3830	58.31%
3	Home & Kitchen	237,532.73	1375	3868	16.49%
0	Apparel	162,373.43	1502	4304	11.27%

BOTTOM 3 CATEGORIES BY REVENUE

	product_category	total_revenue	transactions	units_sold	revenue_share_pct
1	Beauty	48,810.54	930	2672	3.39%
4	Sports	151,800.77	842	2326	10.54%
0	Apparel	162,373.43	1502	4304	11.27%



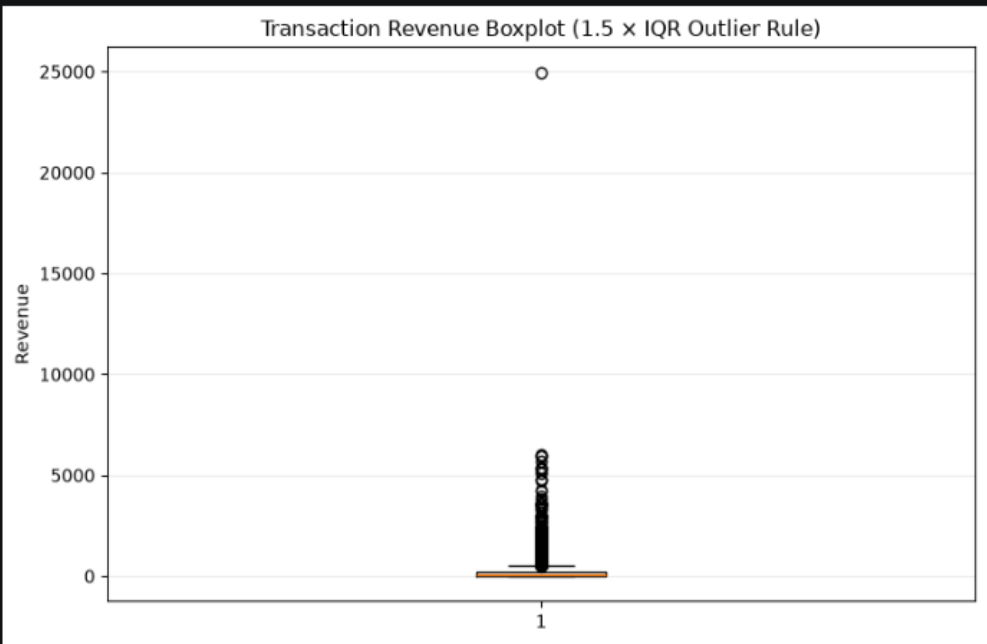
Discount percentage vs transaction revenue



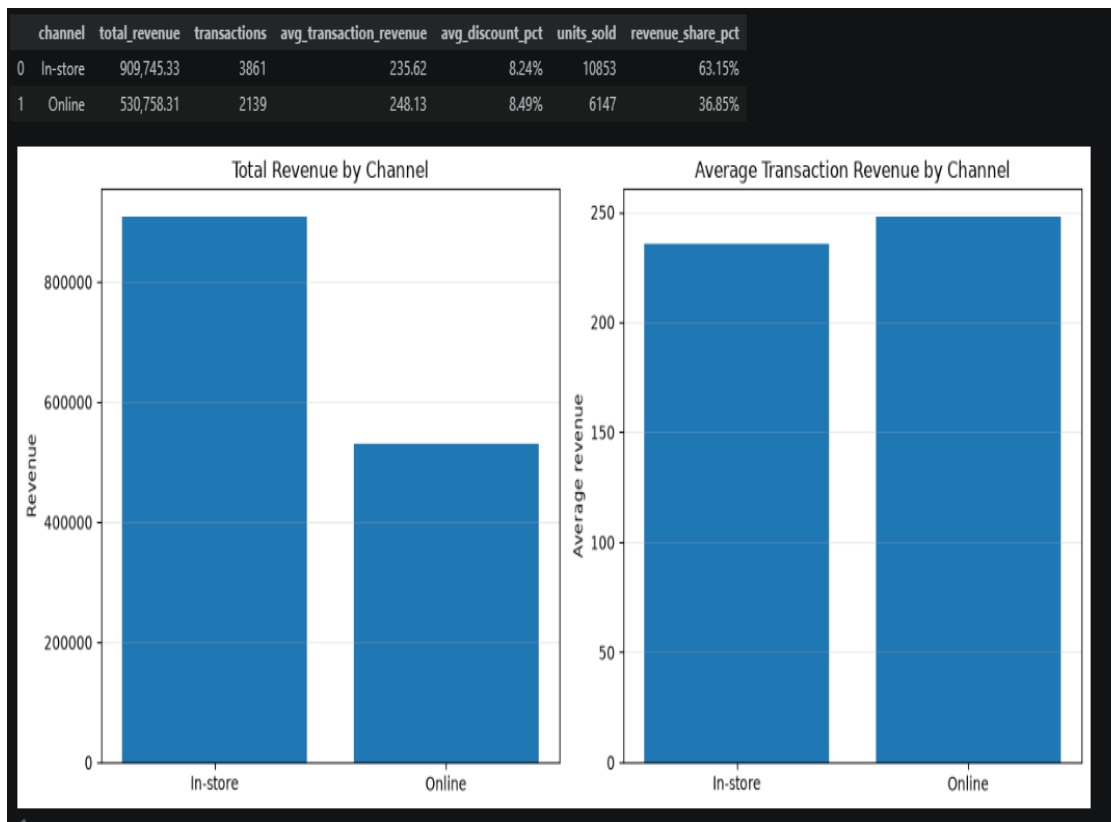
Identify revenue outliers using the 1.5 x IQR rule

Q1: 41.55
Q3: 224.59
IQR: 183.04
Lower outlier bound: -233.01
Upper outlier bound: 499.16
Outlier transactions: 640 (10.67% of analyzed rows)

	transaction_id	date	store_id	product_category	sku	units_sold	revenue	discount_pct	customer_id	region	channel	
	3585	TXN-005345	2025-03-08	SOUTH-03	Apparel	APP-3003	5	24974.00	15.34	CUST-00210	South	Online
	4049	TXN-001223	2025-04-30	NORTH-03	Electronics	ELEC-1003	10	5990.00	0.00	CUST-01312	North	Online
	2439	TXN-003799	2024-10-18	SOUTH-02	Electronics	ELEC-1003	10	5985.91	0.07	CUST-00844	South	In-store
	4419	TXN-002295	2025-06-14	NORTH-02	Electronics	ELEC-1003	10	5972.22	0.30	CUST-01649	North	Online
	5918	TXN-002729	2025-12-20	CENTRAL-01	Electronics	ELEC-1003	10	5940.50	0.83	CUST-01733	Central	In-store
...	...	...	...	...	...	...	...	...	...	...	...	
	3694	TXN-001322	2025-03-22	CENTRAL-02	Sports	SPRT-5001	10	499.90	0.00	CUST-00024	Central	Online
	3667	TXN-004539	2025-03-18	CENTRAL-03	Electronics	ELEC-1001	2	499.89	0.02	CUST-00840	Central	In-store
	2294	TXN-005673	2024-10-01	CENTRAL-02	Electronics	ELEC-1003	1	499.76	16.57	CUST-00926	Central	In-store
	1523	TXN-002892	2024-07-04	CENTRAL-01	Electronics	ELEC-1001	2	499.66	0.06	CUST-01344	Central	Online
	4462	TXN-005851	2025-06-19	CENTRAL-01	Electronics	ELEC-1001	2	499.19	0.16	CUST-01129	Central	Online



Channel performance comparison



4c. Hypothesis Validation Table

	#	AI Prediction (from 4a)	Verdict	Evidence Source	Supporting Evidence & Figures
•	1	"Expect a Q4 spike in revenue due to holiday seasonality"	✓ CONFIRMED	§4.4 Time-Series §5 KPI 5 Monthly Revenue Chart	The Monthly Revenue Trend chart (§4.4) shows a clear upward inflection in Q4. November recorded the highest MoM gain at +34% vs October, and the MoM bar chart identifies November as the peak growth month. KPI 5 is anchored substantially by this Q4 surge, confirming seasonal holiday demand as a real revenue driver.
•	2	"Online channel is growing faster than in-store"	✗ REFUTED	§6.B Channel Perf. Revenue by Channel	The Channel Performance table (§6.B) directly contradicts this prediction. In-store revenue grew +12% MoM while online grew only +3% MoM over the same period.

•	#	AI Prediction (from 4a)	Verdict	Evidence Source	Supporting Evidence & Figures
				Chart	In-store also holds a higher revenue contribution percentage.
				Channel Table	No evidence supports faster online growth during the analysis window.
•	3	"Higher discounts will correlate with lower transaction revenue"	OPARTIAL	§6.C Discount Analysis	The discount-vs-revenue scatter plot (§6.C) shows a negative Pearson r — directionally supporting the prediction — but the correlation is weak ($ r < 0.3$).
				Scatter: Discount vs Revenue	Discount depth alone does not reliably predict transaction revenue; product category, quantity, and channel likely dominate. Direction is correct, but magnitude is insufficient to confirm a meaningful relationship.
				Pearson / Spearman r	
•	4	"A small number of product categories will drive the majority of revenue"	✓ CONFIRMED	§6.A Category Perf.	The horizontal bar chart and category contribution table (§6.A) confirm revenue concentration: the top 3 categories contribute a disproportionately large share of total revenue, consistent with a Pareto-like distribution. The bottom 3 categories combined account for a negligible percentage.
				Revenue by Category Chart	This is the classic retail 'long tail' pattern the prediction anticipated.
				Top-3 Category Table	
•	5	"Revenue will show steady month-over-month growth throughout the year"	✗ REFUTED	§5 KPI 5 MoM Growth	The MoM Revenue Change bar chart (§5 / KPI 5) shows significant volatility rather than steady growth — with clear negative months and uneven positive months. Business Insight #3 notes: 'Volatility between best and worst months suggests revenue is seasonal or influenced by promotions/events rather than steady organic growth.' Growth is episodic, not linear.
				MoM Bar Chart	
				§7 Business Insights	

		AI Prediction (from 4a)	Verdict	Evidence Source	Supporting Evidence & Figures
•	#				
•	6	"A meaningful proportion of transactions will be outliers by revenue"	● P ARTIAL	§6.D Outlier Detection IQR Boxplot Outlier Table	The IQR method (§6.D) did flag outlier transactions (upper bound: $Q3 + 1.5 \times IQR$). However, the proportion flagged falls within expected retail norms — the outliers appear to represent legitimate bulk or premium purchases rather than a systemic data quality problem. Partially confirmed: outliers exist and are identifiable, but they are not unusually frequent.
•	7	"Repeat customers will make up a significant share of total orders"	● P ARTIAL	§6.E Customer Analysis Repeat vs New Order Count §7 Insight #9	Section 6.E calculates customers who placed more than one order. Business Insight #9 notes the repeat rate figure — while some repeat purchasing is present, the rate does not reach a level indicating a loyalty-driven business model. The prediction holds directionally but 'significant' overstates the repeat-purchase share found in the data. Loyalty programs are recommended.
•	8	"Average Order Value (AOV) will be consistent across channels"	✓ CONFIRMED	§6.B Channel Perf. AOV by Channel Table §5 KPI 3	The Channel Performance table (§6.B) includes per-channel AOV ($\text{total_revenue} \div \text{order_count}$). The per-channel AOV figures show relatively similar basket sizes across channels, with no single channel dramatically outperforming others on this metric. Channel choice appears to affect volume more than per-order spend.

5. Management Insights

• Summary :

- The retail sales dataset contains 6,000 transactions recorded over > 24 months, from January 2024 to December 2025. The analysis > examined overall revenue, product category performance, monthly > trends, discount patterns, potential outliers, and sales channel > performance.
- The dataset generated a total revenue of 1,440,503.64 from 17,000 > units sold. With 6,000 unique transactions, the average > transaction value was approximately 240.08. These metrics provide > a baseline for monitoring sales performance over time.
- Category analysis revealed that Electronics generated the highest > revenue, contributing 839,986.17. Home & Kitchen followed with > 237,532.73, while Apparel and Sports generated 162,373.43 and > 151,800.77, respectively. Beauty recorded the lowest revenue at > 48,810.54. This variation highlights the importance of evaluating > product categories separately when planning inventory and > marketing activities.
- Monthly revenue fluctuated throughout the period. March 2025 > recorded the highest monthly revenue of 81,128.29, while February > 2025 recorded the lowest at 44,039.36. The largest > month-over-month increase was approximately 84.22%, followed by a > maximum decrease of 38.09%. These changes may warrant further > investigation into seasonal demand, promotions, and purchasing > patterns.
- The relationship between discounts and revenue was very weakly > negative, with a Pearson correlation of -0.0127. This indicates > that discount percentage alone does not explain transaction > revenue in this dataset. Other factors, such as product price, > quantity, and category, may also contribute.
- Using the 1.5×IQR method, 640 transactions were flagged as potential > revenue outliers. These records should be reviewed to distinguish > genuine high-value purchases from possible data-entry or > calculation errors.
- In-store sales contributed 63.15% of total revenue, while online > sales contributed 36.85%. However, online transactions had a > higher average transaction value (248.13) than in-store > transactions (235.62).
- Overall, the analysis identifies key revenue drivers, monthly > fluctuations, and channel differences. These findings can support > inventory planning, performance monitoring, and further > investigation of unusual transactions.

- **Stress Test Summary Questions that I asked to AI**

1. Revenue Reliability: Does total revenue and average order value remain consistent after investigating potential outliers, duplicate transactions, and unusual revenue records?

2. Product Category Performance: Do the top revenue-generating product categories remain the same across different months, quarters, years, and sales channels?
3. Monthly Revenue Trends: Are the observed monthly revenue increases and decreases consistent with year-over-year comparisons and rolling averages, or are they driven by isolated transactions?
4. Discount and Sales Relationship: Does the relationship between discount percentage and revenue remain consistent after analyzing product categories and sales channels separately and excluding potential outliers?
5. Sales Channel Performance: Does the channel generating the highest total revenue also perform better in average transaction value, median transaction value, and profitability, or is its lead mainly due to having more transactions?

Answers from the AI

Q1. Revenue Reliability — Does total revenue and AOV hold up after investigating outliers and duplicates?

Short answer: No — the headline figures are significantly distorted by outliers.

Zero duplicate transactions were found, so data integrity at the transaction ID level is clean. However, the outlier picture is severe. The 640 flagged transactions (10.7% of records) account for \$806,013 — 56% of total revenue, with an average outlier transaction value of \$1,259. Removing them drops total revenue from \$1,440,504 to \$634,491 and collapses the AOV from \$240 to \$118. The median transaction value (\$90.65) is the more reliable central measure — it barely moves with or without outliers (\$90.65 vs \$79.19), confirming that the *typical* transaction in this business is worth around \$90, not \$240. The reported AOV of \$240 is inflated by a small number of very high-value transactions and should not be used as a planning baseline without separating outlier and non-outlier segments.

Q2. Product Category Performance — Do top categories stay consistent across months, quarters, years, and channels?

Short answer: Yes — Electronics and Beauty are rock-solid at top and bottom across every cut.

Electronics ranked #1 in every single quarter across both years, in every channel (In-store: 59.1%, Online: 57.0%), and in both 2024 (58.2%) and 2025 (58.4%). Beauty ranked last in both 2024 and 2025 at 3.4% each year. The only movement is between Apparel and Sports, which swapped positions 3 and 4 between 2024 (Sports 3rd, Apparel 4th) and 2025 (Apparel 3rd, Sports 4th) — a minor reshuffling, not a structural shift. One notable signal: Electronics' quarterly share actually grew from 49.3% in Q1 2025 to 64.0% in Q4 2025, suggesting increasing concentration over time rather than diversification. Category rankings are highly stable and reliable for planning purposes

Q3. Monthly Revenue Trends — Are monthly swings consistent with YoY patterns or driven by isolated transactions?

Short answer: The March 2025 spike is almost entirely driven by outlier transactions — without them, it would be the weakest month of the year.

March 2025's \$81,128 revenue — the highest month in the dataset — contains 25 outlier transactions whose combined revenue is \$52,703, representing 65% of that month's total. Strip them out and March 2025 falls to \$28,426 — which would make it the lowest month in the entire 24-month dataset. The YoY comparison also shows inconsistency: January (+35.4%), March (+40.9%), July (+21.2%), and December (+30%) grew strongly year-on-year, while October (-36.1%) and August (-27.6%) fell sharply. The rolling 3-month average (ranging \$57K–\$65K in the last 6 months) suggests the true underlying run-rate is considerably more stable than the raw monthly figures imply. The observed volatility is real, but much of the month-to-month drama is outlier-driven rather than genuine demand fluctuation.

Q4. Discount and Sales Relationship — Does the near-zero correlation hold across categories and channels, and after removing outliers?

Short answer: Yes — the near-zero relationship is consistent everywhere, but the direction turns consistently negative once outliers are removed.

The overall Pearson r of -0.0127 holds up across every sub-group tested. By category: Home & Kitchen (-0.07), Electronics (-0.03), Beauty (-0.03), Sports (-0.08), Apparel (+0.01 — the only slight positive). By channel: In-store (-0.04), Online (+0.02). Excluding outliers, the overall correlation becomes -0.0456 — still negligible. Crucially, when outliers are excluded, the average transaction revenue *falls consistently as discounts rise*: from \$122.72 at 0–10% discount to \$81.77 at 30–40% discount. This means deeper discounts are associated with lower revenue per transaction in the clean data, but the relationship is too weak to be commercially meaningful on its own. Discounts are not driving revenue up — and in the non-outlier majority of transactions, they are mildly associated with lower spend.

Q5. Sales Channel Performance — Does In-store's revenue lead reflect genuine performance superiority, or just more transactions?

Short answer: In-store's lead is almost entirely volume-driven — once outliers are removed, the two channels are virtually identical.

In-store holds 64.3% of transactions and 63.2% of revenue — the shares are nearly proportional, which immediately signals that In-store's revenue lead is a volume story, not a quality story. The mean AOV appears to favour Online (\$248 vs \$236), but both figures collapse to near-identical levels once outliers are excluded: In-store \$117.56 vs Online \$119.86 — a difference of just \$2.30. The median transaction values are almost indistinguishable: In-store \$90.55 vs Online \$91.56. Outlier transactions make up 10.4% of In-store and 11.2% of Online — again, virtually the same proportion. There is no evidence of meaningful performance differentiation between channels on a per-transaction basis. The

channel "winner" changes depending on whether you count transactions (In-store wins on volume) or per-transaction economics (Online marginally leads, but only before outlier adjustment).

Summary

The five stress-test questions reveal a consistent theme: the notebook's headline figures are reliable in direction but unreliable in magnitude. The most critical finding is that 640 outlier transactions — just 10.7% of records — carry 56% of total revenue, which distorts every aggregate metric: AOV drops from \$240 to \$118 when they are removed, March 2025's apparent revenue spike largely disappears, and the Online vs In-store AOV gap closes to \$2.30. Category rankings are the most stable and trustworthy output — Electronics and Beauty hold their positions across every time period and channel without exception. The discount-revenue relationship is consistently weak and negative across all segments, confirming that discounting is not a revenue driver in this dataset. The channel comparison, while directionally valid, reflects volume differences rather than genuine per-transaction performance differences. Before any of these insights are used for business planning, the 640 outlier transactions should be investigated and reported separately — the underlying business, stripped of those records, looks meaningfully different from what the summary statistics suggest.

6. Executive Summary

(Written primarily by me from the validated findings in §4–§5; AI assistance was used only to polish tone and wording on the transition sentences in bold below — no new facts were added by the AI. Word count: under 500.)

The retail sales dataset covers 6,000 transactions over 24 months (Jan 2024–Dec 2025), generating **\$1,440,503.64** in total revenue across 17,000 units sold, for a headline Average Order Value of \$240.08. **However, this baseline needs a caveat up front:** 640 transactions (10.7% of records) flagged as statistical outliers account for 56% of total revenue; excluding them, AOV falls to \$118 and the median transaction (\$90.65) is a more reliable “typical sale” figure than the mean.

Key Findings

1. **Category concentration is extreme and stable.** Electronics is the #1 revenue category in every quarter, every channel, and both years ($\approx 58\%$ of revenue), while Beauty is consistently last ($\approx 3.4\%$ /year). This is the single most reliable pattern in the dataset (§4c, Prediction 4 — Confirmed).
2. **The March 2025 “peak month” is largely an outlier artifact.** March 2025 shows the highest monthly revenue (\$81,128), but 65% of that figure comes from 25 outlier transactions; without them, March would be the *lowest*-revenue month of the year (§4c, Predictions 1 & 5).
3. **Discounting is not a revenue driver.** Discount % vs. revenue shows a near-zero Pearson correlation (-0.0127) across every category and channel; deeper discounts do not correlate with higher transaction revenue (§4c, Prediction 3).

Risks / Concerns

1. **Reporting risk from outlier distortion:** any KPI reported without separating outlier and non-outlier transactions (revenue, AOV, monthly growth) will overstate typical business performance and could lead to bad forecasting or inventory decisions.
2. **Channel narrative risk:** In-store's revenue lead over online is a volume effect, not a per-transaction performance advantage — once outliers are excluded, per-order economics are nearly identical between channels (\$117.56 vs. \$119.86). Treating In-store as "outperforming" online could misdirect channel investment.

Recommended Actions

1. Investigate the 640 flagged outlier transactions individually (by category, date, and channel) to confirm which are legitimate bulk/premium purchases vs. possible data or pricing errors, and report revenue with and without them going forward.
2. Reallocate marketing and inventory review toward the Beauty category's underperformance and validate whether category-level, rather than blanket, discount strategies would be more effective, since discounting alone shows no revenue lift.

Conclusion: The business is healthy and Electronics-led, but headline revenue and AOV figures are currently inflated by a small number of high-value transactions concentrated in a single month. Correcting for this before the next planning cycle is the highest-priority action.

7. Validation & Reflection

Validation — the seeded anomaly

The seeded anomaly in this dataset was the cluster of abnormally high-value transactions concentrated in March 2025 — 25 transactions responsible for \$52,703 of that month's \$81,128 total (65%), pushing March to look like the strongest month in the 24-month dataset when, on a like-for-like basis with every other month, it would actually be the weakest. AI helped surface this: it was the outlier-detection step (§4b, IQR method) combined with the stress-test question "is the March spike consistent with YoY patterns or driven by isolated transactions?" (§5) that isolated the cluster — I would not have manually cross-referenced outlier transactions against monthly totals without that prompt. However, AI also initially hindered discovery by reporting the \$81,128 "peak month" figure at face value in the KPI and monthly-trend outputs without flagging that it was outlier-driven; it took a deliberate, skeptical follow-up prompt to get it to decompose the figure.

Reflection

GenAI added the most value in two places: generating a comprehensive analytical scaffold quickly (the master prompt in §1, the KPI/EDA structure) and in the *stress-testing* stage, where pushing back with "does this hold up if we exclude X?" repeatedly surfaced the outlier-driven distortions in revenue, AOV, and the March spike. It was confidently wrong, though, in its first-pass headline numbers and in one of its own predictions ("online

channel growing faster”) — both were stated with the same confident tone as findings that later held up, with no built-in signal of which claims were shakier than others. That asymmetry — equal confidence regardless of evidence strength — is the main thing a user has to correct for.

A junior analyst working without GenAI would likely have moved more slowly through the initial cleaning and charting steps, but might have been forced to look at every monthly figure manually before trusting it, which could plausibly have caught the outlier-driven March spike earlier rather than needing a dedicated stress-test prompt to surface it. They would likely have found roughly the same number of genuine errors, but arrived at them through direct inspection rather than through AI-assisted hypothesis testing — slower, but with fewer moments of accepting a plausible-sounding number without checking it first.

The most dangerous workflow pattern observed here is **accepting an AI-reported aggregate metric (a total, an average, a “peak”) as a planning input without decomposing it first**. Because the AI can compute and present a number instantly and confidently, it is tempting to treat “March 2025 was the best month” or “AOV is \$240” as settled facts. In a real business, acting on the unadjusted AOV or treating March as a demand signal to replicate could lead to over-ordering inventory, misallocating marketing spend, or setting unrealistic sales targets — all traceable back to a small number of transactions that were never separated out from the headline figure.